THE ALGEBRAIC INDUCTION STEP: FLATNESS OF A LOCALISED MODULE OVER A FLAT FINITELY PRESENTED RING MAP WITH GEOMETRICALLY INTEGRAL FIBRES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Universal injectivity over a local base for a map whose reduction modulo the maximal ideal is injective. Same chapter flat.tex, labels lemma-universally-injective-local (proof line 2396) and lemma-fibres-irreducible-flat-projective-nonnoetherian (proof line 2397). Together they show that the map from the free module to the localised module is universally injective over R once its reduction modulo the maximal ideal is injective, which is the heart of the implication (1) implies (2).
- Permanence and localisation of universal injectivity, and flatness of the cokernel of a universally injective map over a domain. Algebra, labels algebra-lemma-universally-injective-permanence (proof line 2400), algebra-lemma-universally-injective-localize (proof line 2403) and algebra-lemma-ui-flat-domain (proof line 2405). Carries universal injectivity from the localised map back to the map on the local ring and makes the cokernel flat over R, which produces condition (2).
- Nakayama's lemma, nonzerodivisors of a flat finitely presented ring map, and flatness of an extension. Algebra, labels algebra-lemma-NAK (proof line 2411); Same chapter flat.tex, labels lemma-invert-universally-injective (proof line 2417) and lemma-fibres-irreducible-flat-projective-nonnoetherian (proof line 2418); Algebra, label algebra-lemma-flat-ses (proof line 2421). Nakayama upgrades injectivity at p to an isomorphism, the nonzerodivisor statement lets injectivity at p be pulled back to injectivity at q, and flatness of an extension closes the cycle from (5) back to (1).

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Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Navigation pointer for \ref{algebra-lemma-NAK} used inside the counted span (line 2411); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HM: sources/stacks-chow/tags.txt, source lines 648–648. Navigation pointer for \ref{algebra-lemma-flat-ses} used inside the counted span (line 2421); the argument that consumes it is retained in the span.

Source reference. Original lemma 058P: sources/stacks-chow/tags.txt, source lines 6787–6787. Navigation pointer for \ref{algebra-lemma-ui-flat-domain} used inside the counted span (line 2405); the argument that consumes it is retained in the span.

Source reference. Original lemma 05CM: sources/stacks-chow/tags.txt, source lines 6925–6925. Navigation pointer for \ref{algebra-lemma-universally-injective-localize} used inside the counted span (line 2403); the argument that consumes it is retained in the span.

Source reference. Original lemma 05CJ: sources/stacks-chow/tags.txt, source lines 6922–6922. Navigation pointer for \ref{algebra-lemma-universally-injective-permanence} used inside the counted span (line 2400); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FT: sources/stacks-flat/pinned-original-source.tex, source lines 2247–2247. Navigation pointer for \ref{lemma-fibres-irreducible-flat-projective-nonnoetherian} used inside the counted span (line 2397); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FR: sources/stacks-flat/pinned-original-source.tex, source lines 1976–1976. Navigation pointer for \ref{lemma-invert-universally-injective} used inside the counted span (line 2417); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FQ: sources/stacks-flat/pinned-original-source.tex, source lines 1946–1946. Navigation pointer for \ref{lemma-universally-injective-local} used inside the counted span (line 2396); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $(R, \mathfrak{m})$ be a local ring. Let $R \to S$ be a finitely presented flat ring map with geometrically integral fibres. Write $\mathfrak{p} = \mathfrak{m}S$. Let $\mathfrak{q} \subset S$ be a prime ideal lying over $\mathfrak{m}$. Let N be a finite S-module. There exist $r \geq 0$ and an S-module map*

$$\alpha : S^{\oplus r} \longrightarrow N$$

such that $\alpha : \kappa(\mathfrak{p})^{\oplus r} \to N \otimes_S \kappa(\mathfrak{p})$ is an isomorphism. For any such α the following are equivalent:

1. *$N_\mathfrak{q}$ is R-flat,*
2. *α is R-universally injective and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat,*
3. *α is injective and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat,*
4. *$\alpha_\mathfrak{p}$ is an isomorphism and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat, and*
5. *$\alpha_\mathfrak{q}$ is injective and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat.*

Proof. To obtain α set $r = \dim_{\kappa(\mathfrak{p})} N \otimes_S \kappa(\mathfrak{p})$ and pick $x_1, \ldots, x_r \in N$ which form a basis of $N \otimes_S \kappa(\mathfrak{p})$. Define $\alpha(s_1, \ldots, s_r) = \sum s_i x_i$. This proves the existence.

Fix an α. The most interesting implication is (1) $\Rightarrow$ (2) which we prove first. Assume (1). Because $S/\mathfrak{m}S$ is a domain with fraction field $\kappa(\mathfrak{p})$ we see that $(S/\mathfrak{m}S)^{\oplus r} \to N_\mathfrak{p}/\mathfrak{m}N_\mathfrak{p} = N \otimes_S \kappa(\mathfrak{p})$ is injective. Hence by Lemmas 05FQ and 05FT. the map $S^{\oplus r} \to N_\mathfrak{p}$ is R-universally injective. It follows that $S^{\oplus r} \to N$ is R-universally injective, see Algebra, Lemma 05CJ. Then also the localization $\alpha_\mathfrak{q}$ is

R-universally injective, see Algebra, Lemma 05CM. We conclude that $\mathrm{Coker}(\alpha)_{\mathfrak{q}}$ is R-flat by Algebra, Lemma 058P.

The implication (2) $\Rightarrow$ (3) is immediate. If (3) holds, then $\alpha_{\mathfrak{p}}$ is injective as a localization of an injective module map. By Nakayama's lemma (Algebra, Lemma 00DV) $\alpha_{\mathfrak{p}}$ is surjective too. Hence (3) $\Rightarrow$ (4). If (4) holds, then $\alpha_{\mathfrak{p}}$ is an isomorphism, so α is injective as $S_{\mathfrak{q}} \to S_{\mathfrak{p}}$ is injective. Namely, elements of $S \backslash \mathfrak{p}$ are nonzerodivisors on S by a combination of Lemmas 05FR and 05FT. Hence (4) $\Rightarrow$ (5). Finally, if (5) holds, then $N_{\mathfrak{q}}$ is R-flat as an extension of flat modules, see Algebra, Lemma 00HM. Hence (5) $\Rightarrow$ (1) and the proof is finished. $\square$